Data Quality Report

Dataset Reviewed

Merged county-level dataset combining EPA Annual AQI by County (2026) and CDC PLACES County Asthma Data (2025 release), joined on standardized State + County identifiers.

Quality Checklist Results

Check	Result	Issue Found?	Action Taken
Missing values	187 counties missing asthma data; some counties missing population after outer join	Yes	Filled asthma gaps with column mean via np.where(); kept population as nullable Int64 rather than dropping rows
Data types	TotalPopulation stored as comma-formatted text; CI values stored as text	Yes	Converted population to integer; split CI text into numeric ci_lower/ci_upper columns
Duplicate rows	6 State+County combos appeared twice (e.g., Baltimore city vs. county)	Yes (not true duplicates)	Identified as independent cities vs. counties; used CountyFIPS as the true unique key instead of dropping rows
Outliers	Checked asthma prevalence range and Median AQI range	No	All values fell within realistic bounds (0-100% and non-negative AQI); no capping needed
Inconsistent categories	State names: full names vs. abbreviations across the two files	Yes	Standardized casing/format on both State columns before merging
Formatting issues	Checked for extra whitespace in county/state name fields	Minor	Applied .str.strip() to name columns as a precaution before joining

Findings Summary

- Missing data was the most significant issue: 187 of 3,143 counties (5.9%) had no asthma estimate, and an unknown number of counties had no AQI monitor at all, meaning they only appear on one side of the merge.
- The county 'duplicates' were a false alarm on first glance — investigating them revealed they were legitimate distinct places (an independent city and a same-named surrounding county), not data errors. This is an important reminder that duplicate() flags need manual review, not automatic deletion.
- No genuine outliers were found in either core numeric column (asthma prevalence, Median AQI) — both datasets appear to already enforce realistic value ranges at the source (CDC/EPA), unlike hand-entered data which is more outlier-prone.
- Text formatting and category inconsistency were minor and mostly preventative fixes (standardizing state name format) rather than fixes for problems that had already caused visible errors.

Final Validation Status

- No missing values remain unaddressed — remaining gaps are explicitly flagged (has_asthma_data, has_aqi_data) rather than silently filled or dropped.
- Data types are correct — population is integer, CI bounds are numeric, state formats match across both files.
- No unresolved duplicate rows — the six same-named counties are confirmed as distinct records using CountyFIPS.
- No extreme outliers distort the analysis.
- Text data is formatted consistently across both source files.

The dataset passes all five checklist items and is ready for further analysis.